

# Qi Sun

New York, NY • [qs2196@nyu.edu](mailto:qs2196@nyu.edu)

## EDUCATION

New York University Shanghai

B.S. in Computer Science

Cumulative GPA: 3.638 / 4.000

Shanghai, China

Expected May 2027

## PUBLICATIONS

1. **Qi Sun**, Siyue Zhang, Yulin Chen, Yuxiang Xue, Ru Peng, Chen Zhao. “From ‘Weak’ Signals to Strong Models: Preference Delta Aggregation with LoRA Merging.” *Submitted to NeurIPS 2026 (under review)*. [arXiv:2606.00357](https://arxiv.org/abs/2606.00357).
2. Yizhou Liu\*, **Qi Sun\***, Yulin Chen, Siyue Zhang, Chen Zhao. “Search, Do not Guess: Teaching Small Language Models to Be Effective Search Agents.” *ARR (Jan 2026)*; meta-review accepted to Findings; in resubmission to EMNLP 2026. [arXiv:2604.04651](https://arxiv.org/abs/2604.04651).

\*Equal contribution.

## RESEARCH EXPERIENCE

### Research Assistant

2026.05 – Present

Advisor: Prof. Artie Shen

NYU Langone Health, New York, NY

- **Project:** Feature-Oriented Multimodal Retrieval for Early Pancreatic Cancer (PDAC) Detection
- Design a feature-oriented retrieval system over abdominal CT/MRI radiology reports and structured EHR to surface diagnostically similar cases for ambiguous pancreatic presentations.
- Build a clinical codebook that encodes radiographic findings, quantitative thresholds, and negation phrasings as structured features driving a hybrid BM25 + dense retrieval index.
- Generalize the system into a disease-agnostic “cartridge” configuration, enabling reuse across conditions without re-engineering the core retrieval pipeline.
- Integrate retrieval with internal LLMs to estimate malignancy probability and generate evidence-grounded differential diagnoses.

### Research Project

2026.06 – Present

Advisor: Prof. Chen Zhao

NYU Shanghai

- **Project:** Distribution-Aware Offline On-Policy Distillation (in progress)
- Investigate distribution-fixation methods extending Lightning On-Policy Distillation (OPD) to the offline setting, targeting more stable distillation of agentic-search behavior.
- Study how distributional mismatch between teacher rollouts and student policy degrades offline distillation, and how to correct it without online sampling.

### Research Project

2026.01 – 2026.05

Advisor: Prof. Chen Zhao

NYU Shanghai

- **Project:** Preference Delta Aggregation (PDA) for Weak-to-Strong LLM Training
- Proposed PDA, a framework that converts each weak-weaker model pair (e.g., Qwen3 4B over 1.7B) into a preference delta, instantiated as a LoRA adapter and aggregated via LoRA merging.
- Introduced Geometric Alignment Merging (GAM) to align adapter subspaces and reduce directional interference when merging diverse deltas.
- Showed that aggregating multiple weak signals improves a strong Qwen3-8B model by 6.8 / 7.3 points on knowledge-reasoning and agentic-search benchmarks, exceeding the best single-delta baseline by 2.1 / 4.3 points.
- First author; submitted to NeurIPS 2026.

### Research Project (Dean’s Undergraduate Summer Research Funding)

2025.06 – 2025.12

Advisor: Prof. Chen Zhao

NYU Shanghai

- **Project:** Search, Do not Guess — Teaching Small Language Models to Be Effective Search Agents
- Reproduced and benchmarked agentic-search baselines (Search-O1, Search-R1) for controlled comparison.
- Distilled agentic-search behavior from Qwen3-32B into Qwen3-1.7B using SFT and on-policy distillation.
- Designed probing and rejection fine-tuning techniques to teach adaptive (when-to-search) behavior, improving small-model search effectiveness on multi-hop reasoning tasks.
- Co-first author; accepted to Findings via ARR meta-review, in resubmission to EMNLP 2026.

### Research Assistant

2025.02 – 2025.05

Advisor: Prof. Xianbin Gu

NYU Shanghai

- **Project:** LLM-Enhanced ICPC Training Platform
- Developed a full-stack training platform (Vue3, Django) with an online judge for the NYU Shanghai ICPC team.
- Engineered an LLM-based recommendation system that generates personalized learning roadmaps and problem suggestions from user progress.

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| <b>Research Project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024.08 – 2025.05 |
| <i>Advisor: Prof. Chen Zhao</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NYU Shanghai      |
| <ul style="list-style-type: none"> <li>• <b>Project:</b> Open-Domain Fact-Checking with “Not Enough Information” (NEI) Labels</li> <li>• Analyzed NEI label distribution and dataset shift across fact-checking benchmarks (VitaminC, SciFact).</li> <li>• Ran NLP baselines and manual error analysis to characterize common failure modes in NEI classification.</li> <li>• Fine-tuned open-domain models for more robust text classification under ambiguous-evidence settings.</li> </ul> |                   |
| <b>Research Project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024.06 – 2024.08 |
| <i>Advisor: Prof. Chen Zhao</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NYU Shanghai      |

- **Project:** Financial Reports Analyzing System with NLP
- Built a system to summarize 10-K and 10-Q financial reports using BART, LLaMA, and GPT models.
- Implemented an MPNet-based retrieval method to trace generated summaries back to their source text.

## TECHNICAL SKILLS

- **Languages:** Python, C/C++ , Java, R, JavaScript, HTML/CSS, SQL
- **Frameworks & Tools:** PyTorch, Transformers, PEFT/LoRA, LangChain, FAISS, Pandas, NumPy, Git, Flask, Django, Vue3, React, Next.js
- **Research Areas:** Agentic Retrieval-Augmented Generation (RAG), Preference Optimization, LoRA Merging, Knowledge Distillation, Fine-Tuning (SFT, RLHF), Multimodal Medical Retrieval

## LEADERSHIP & COMPETITION

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|-------------------------------------------------------------------------|-------------------|
| • <b>Chair</b> , NYU Shanghai ACM Student Chapter                       | 2024.06 – 2025.06 |
| • <b>Director of Competition</b> , NYU Shanghai ICPC Training Community | 2024.06 – 2025.06 |
| • <b>ICPC Regional Contest, Ningxia</b> — Bronze Prize                  | 2025.06           |
| • <b>CCPC Regional Contest, Shanghai</b> — Bronze Prize                 | 2024.06           |